Institute of Technology of Cambodia

Department of Information and Communication Engineering

Assignment: Lab04

Subject: **Operating System**

Lecturer Course/TP: **Heng Rathpisey**

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Year: **I4**

Academic year: 2024-2025

1. **List contents of the current directory, including ownership and permissions, and redirect to** contents.txt **in the home directory.**

```
sokmeak@Labtop-SokmeakSR:~$ ls -l
total 60
drwxr-xr-x 2 sokmeak sokmeak  4096 Nov 29 22:06 CopyingFolder
drwxr-xr-x 5 sokmeak sokmeak  4096 Nov 29 23:02 Desktop
drwxr-xr-x 4 sokmeak sokmeak  4096 Nov 29 23:08 Document
drwxr-xr-x 3 sokmeak sokmeak  4096 Nov  5 10:48 LabFiles
drwxr-xr-x 7 sokmeak sokmeak  4096 Nov  8 13:46 OS2024
-rwxr-xr-x 1 sokmeak sokmeak 15952 Oct 25 14:35 hello
-rw-r--r-- 1 sokmeak sokmeak   248 Oct 25 14:32 hello.c
drwxr-xr-x 4 sokmeak sokmeak  4096 Oct 22 13:58 ic-project
drwxr-xr-x 2 sokmeak sokmeak  4096 Nov 29 23:59 link_lab
drwxr-xr-x 2 sokmeak sokmeak  4096 Nov 29 23:51 logs
drwxr-xr-x 3 sokmeak sokmeak  4096 Oct 25 13:53 myfolder
drwxr-xr-x 3 sokmeak sokmeak  4096 Nov 29 23:44 wildcard_lab
sokmeak@Labtop-SokmeakSR:~$ ls -l > contents.txt
sokmeak@Labtop-SokmeakSR:~$ ls
CopyingFolder  LabFiles      hello       link_lab  wildcard_lab
Desktop        OS2024        hello.c     logs
Document       contents.txt  ic-project  myfolder
sokmeak@Labtop-SokmeakSR:~$ cat contents.txt
```

2. Sort the contents of contents.txt and append to contents-sorted.txt.

```
sokmeak@Labtop-SokmeakSR:~$ sort ~/contents.txt >> ~/contents-sor
ted.txt
sokmeak@Labtop-SokmeakSR:~$ cat contents-sorted.txt
-rw-r--r-- 1 sokmeak sokmeak     0 Nov 30 22:49 contents.txt
-rw-r--r-- 1 sokmeak sokmeak   248 Oct 25 14:32 hello.c
-rwxr-xr-x 1 sokmeak sokmeak 15952 Oct 25 14:35 hello
drwxr-xr-x 2 sokmeak sokmeak  4096 Nov 29 22:06 CopyingFolder
drwxr-xr-x 2 sokmeak sokmeak  4096 Nov 29 23:51 logs
drwxr-xr-x 2 sokmeak sokmeak  4096 Nov 29 23:59 link_lab
drwxr-xr-x 3 sokmeak sokmeak  4096 Nov  5 10:48 LabFiles
drwxr-xr-x 3 sokmeak sokmeak  4096 Nov 29 23:44 wildcard_lab
drwxr-xr-x 3 sokmeak sokmeak  4096 Oct 25 13:53 myfolder
drwxr-xr-x 4 sokmeak sokmeak  4096 Nov 29 23:08 Document
drwxr-xr-x 4 sokmeak sokmeak  4096 Oct 22 13:58 ic-project
drwxr-xr-x 5 sokmeak sokmeak  4096 Nov 29 23:02 Desktop
drwxr-xr-x 7 sokmeak sokmeak  4096 Nov  8 13:46 OS2024
total 60
sokmeak@Labtop-SokmeakSR:~$
```

3. Display the last 10 lines of **/etc/passwd** and redirect to a file in the **Documents** directory.

```
sokmeak@Labtop-SokmeakSR:~$ tail -n 10 /etc/passwd > ~/Document/passwd_tail.txt
sokmeak@Labtop-SokmeakSR:~$ cat ~/D
Desktop/  Document/
sokmeak@Labtop-SokmeakSR:~$ cat ~/Document/passwd_tail.txt
systemd-network:x:998:998:systemd Network Management:/:/usr/sbin/nologin
systemd-timesync:x:996:996:systemd Time Synchronization:/:/usr/sbin/nologin
dhcpcd:x:100:65534:DHCP Client Daemon,,,:/usr/lib/dhcpcd:/bin/false
messagebus:x:101:101::/nonexistent:/usr/sbin/nologin
syslog:x:102:102::/nonexistent:/usr/sbin/nologin
systemd-resolve:x:991:991:systemd Resolver:/:/usr/sbin/nologin
uuidd:x:103:103::/run/uuidd:/usr/sbin/nologin
landscape:x:104:105::/var/lib/landscape:/usr/sbin/nologin
polkitd:x:990:990:User for polkitd:/:/usr/sbin/nologin
sokmeak:x:1000:1000:,,,:/home/sokmeak:/bin/bash
sokmeak@Labtop-SokmeakSR:~$
```

4. Count the number of words in **contents.txt** and append the output to **field2.txt** in the home directory.

```
sokmeak@Labtop-SokmeakSR:~$ wc -w < ~/contents.txt >> ~/field2.tx
t
sokmeak@Labtop-SokmeakSR:~$ cat ~/field2.txt
119
sokmeak@Labtop-SokmeakSR:~$
```

5. Display the first 5 lines of /etc/passwd and sort them in reverse alphabetical order.

```
sokmeak@Labtop-SokmeakSR:~$ head -n 5 /etc/passwd | sort -r
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
```

6. Count the number of characters in the last 9 lines of **contents.txt**.

```
sokmeak@Labtop-SokmeakSR:~$ cat ~/contents.txt
total 60
drwxr-xr-x 2 sokmeak sokmeak  4096 Nov 29 22:06 CopyingFolder
drwxr-xr-x 5 sokmeak sokmeak  4096 Nov 29 23:02 Desktop
drwxr-xr-x 4 sokmeak sokmeak  4096 Nov 29 23:08 Document
drwxr-xr-x 3 sokmeak sokmeak  4096 Nov  5 10:48 LabFiles
drwxr-xr-x 7 sokmeak sokmeak  4096 Nov  8 13:46 OS2024
-rw-r--r-- 1 sokmeak sokmeak     0 Nov 30 22:49 contents.txt
-rwxr-xr-x 1 sokmeak sokmeak 15952 Oct 25 14:35 hello
-rw-r--r-- 1 sokmeak sokmeak   248 Oct 25 14:32 hello.c
drwxr-xr-x 4 sokmeak sokmeak  4096 Oct 22 13:58 ic-project
drwxr-xr-x 2 sokmeak sokmeak  4096 Nov 29 23:59 link_lab
drwxr-xr-x 2 sokmeak sokmeak  4096 Nov 29 23:51 logs
drwxr-xr-x 3 sokmeak sokmeak  4096 Oct 25 13:53 myfolder
drwxr-xr-x 3 sokmeak sokmeak  4096 Nov 29 23:44 wildcard_lab
sokmeak@Labtop-SokmeakSR:~$ head -n 9 ~/contents.txt | wc -m
467
sokmeak@Labtop-SokmeakSR:~$
```

7. Count the number of files named **test** in **/usr/share** and its subdirectories.

```
sokmeak@Labtop-SokmeakSR:~$ ls /usr/share
GConf            console-setup        drirc.d      info               man-db        polkit-1                  tabset
PackageKit       consolefonts         file         info.dir           maven-repo    profile                   terminfo
X11              consoletrans         fish         initramfs-tools    menu          profile.md5sums           test.txt
applications     dbus-1               fontconfig   installed-tests    metainfo      publicsuffix              themes
apport           debconf              fonts        iso-codes          mime          python-apt                thumbnailers
appstream        debhelper            gcc          java               misc          python-babel-localedata   ubuntu-release-upgrader
awk              debianutils          gdb          keyrings           motd          python3                   ucf
base-files       defaults             gettext      landscape          nano          readline                  unattended-upgrades
base-passwd      dhcpcd               git-core     libc-bin           netplan       rsync                     upstart
bash-completion  dict                 gitweb       libdrm             networks      rsyslog                   util-linux
binfmts          distro-info          glib-2.0     libgcrypt20        pam           sensible-utils            vim
bug              doc                  glvnd        libthai            pam-configs   session-migration         vulkan
build-essential  doc-base             gnupg        lintian            pastebin.d    sgml                      xml
byobu            dot.bashrc           groff        locale             perl          sgml-base                 xml-core
ca-certificates  dot.profile          gtk-3.0      locales            perl5         snapd                     zoneinfo
cmake            dot.profile.md5sums  i18n         lto-disabled-list  pixmaps       staff-group-for-usr-local zsh
common-licenses  dpkg                 icons        man                pkgconfig     systemd
```

```
sokmeak@Labtop-SokmeakSR:~$ find /usr/share -type f -name "test*" | wc -l
35
sokmeak@Labtop-SokmeakSR:~$ find /usr/share -type f -name "test.txt" | wc -l
1
sokmeak@Labtop-SokmeakSR:~$
```